UNSW Business School/
Information Systems and
Technology Management

INFS5831/4831 T1 2019
Individual Assignment

Individual Assignment Cover Sheet

Please enter your details into the areas below.

Course Name:	**Information Systems Consulting**	**Course ID: INFS5831/4831**
Due Date	**Friday, 29th March 2019** (electronic submission)	
Student ID:	**z5186243**	
Surname:	**Aslam**	
Given Name:	**Hashim Ahmed**	
Student Email:	**hashimahmed.aslam@student.unsw.edu.au**	
Assignment Word Length: 2494		**Not to exceed 2,500 words**

By completing the above form and submitting this assignment, I certify that this assignment is my own work, based on my personal study and/or research, and that I have acknowledged all material and sources in the preparation of this assignment whether they be books, articles, reports, lecture notes, any other kind of document or personal communication.

I also certify that the assignment has not previously been submitted for assessment in any other course or at any other time in the same course and that I have not copied in part or whole or otherwise plagiarised the work of other students and/or persons.

Signature: **STUDENT :**	**Date:** 29th March 2019

Client Needs Analysis

Requirement Analysis

This SAS–ERP change management project entails consideration of multiple factors that will influence the transformation of Queensland Health's incumbent 20-year-old SAP system to the cloud-based SAP HANNA version. The client requires a competent and experienced IT consulting firm with established frameworks capable of delivering a complex change management service. It is critical that prospective tender applicants are particularly well-equipped to undertake a project of such scale and have demonstrated experience in SAP financial system transformation, contrary to the retrospective events of transforming LATTICE payroll system with IBM as partner (Government, 2017). One of the key client requirements is to ensure a smooth integration across the finance, logistics, warehousing, assets, and procurement departments into their recently stabilised pay-roll system (Cowan, 2017). As specified by the RFP, the transformation to renew the financial system warrants the implementation of the SAP HANNA platform, establishing a program to upskill users and the development of a prototype SAP HANA software that demonstrate the effectiveness of the expected final system to induce user confidence (Purnell, 2017).

Furthermore, as corroborated by the Queensland audit office in an audit, the Queensland health department' financial system–a 20-year-old SAP system–remained unsupported for more than a decade, which resulted in major instabilities and performance issues of the respective department (Pearce, 2017). The operating system and the database have also become defunct without vendor support and will be difficult to replicate as the new systems may not be compatible (Pearce, 2017). This implies that the transformation must be managed thoroughly, considering all the stakeholders, to ensure absolute compatibility with their existing process flows and database. The client prioritises the new system to be integrated with the contemporary systems, build a support environment system to compliment user experience and easy access to all health systems and tools, devise a secure information sharing across departments and ensure inter-functionality and enterprise system upgrade for finance, logistics, warehousing, assets, procurement departments and patient administration (Health, 2015).

Stakeholders Identification

Table 1 Stakeholder Identification

Stakeholders	Roles	Interests
Health Services Chief Executives • Minister for Health and Ambulance Services • Director-General (Ref)	Prioritisation and planning of the ICT infrastructure implementation process	Establish a well-planned strategy for ERP transformation (Health, 2015)
Hospital and Health Services • CEO • Board Members	ICT infrastructure planning and engagement with HSS	Communication of agreement updates with Queensland Health Department to ensure quality service (Pearce, 2017)
Chief Health Information Officer	Part of ICT planning process	Strategy formation for ERP upgrade (Health, 2015)
Chief Technology Officer	Part of ICT planning process	Strategy formation for ERP upgrade (Health, 2015)
QHEST[1] Team	Provide functional, transition and project management support	Ensure smooth integration of SAP ERP with respective departments (Cowan, 2017)
Queensland Health Employees	Users of the New SAP ERP system	Upskill themselves to operate the new system (Purnell, 2017) Easy access to health systems and tools (Health, 2015) *Take part in testing and provide feedback*
Patients of the HHS[2] network	Service utilisation	Receive uninterrupted and high-quality service with information security (Health, 2015)
FSR[3] Team	Financial system renewal technical support	Collaboration with the consultants to ensure smooth inter-dependent integration,

[1] Queensland Health Enterprise solutions transition team
[2] Hospital and Health services
[3] Financial System Renewal

		particularly finance department
SAS	Owner of SAP S/4HANA	Ensure system compatibility and functionality based on client needs

Engagement Approach

Pair-of-hands role

The transformation of the SAP ERP system can essentially have multiple implications for project stakeholders depending on the way the consultants take charge of the project. In the case of Queensland Health there are up to 75,000 employees that will be directly affected if there are any complications in the transition (Manongdo, ND). Drawing from the multiple system transformation failures Queensland Health has previously suffered, the top management is heavily invested in the implementation of the system into their contemporary processes as the risk associated with the unsuccessful transformation is too catastrophic. Therefore, the consultants may undertake a collaborative role for this engagement as this may be a secure selling point in the case of ERP change management projects (Caldas & Wood, Jr., 1999). However, a collaborative role might undersell the capabilities of the consulting company as they would only provide generalised service which can turn out to be detrimental. Furthermore, undertaking the expert role may induce reluctance at the client end and this approach may seem too aggressive to them (Caldas & Wood, Jr., 1999). The client aims to hire contractors with specialist knowledge in software integration, support and migration (Coyne, 2017). A pair-of-hands role would allow the consultant to act on specific change requirements laid out by the client and provide specialised knowledge in the transformation of the SAP ERP system to ensure smooth integration across the departments highlighted by the Queensland Health department.

Vendor, consultant relationship

The Queensland Health SAP system upgrade and the LATTICE upgrade projects have been prone to arising conflicts of interest. For instance, with their engagement with IBM, the latter attempted to implement a system that was never implemented in Australian public sector and had no connection with SAP as highlighted by the Queensland government (Cowan, 2015). This could mean that IBM had their own personal interest in implementing a software they preferred, or might have a deal with the external provider, and therefore were unable to map that system as per the client requirements.

Consultant (Financially Motivated)

Another source of conflict would be misleading the client by providing false information pertaining to their capabilities to undertake the project. Queensland government claimed that IBM did the same to primarily win the bid for their business interests (Cowan, 2015). The client will still have to pay the consultant if the underperformance terms are not clearly defined.

Employees

Moreover, employees can also be a potential source of conflict of interest as their own ulterior motives could hinder the intended transformation. For instance, they could falsify or exaggerate

feedback which would make the process more complex as the ambiguity of the feedback in general would increase further limiting the ability of the senior management to devise a solution.

Managing Conflicts

One way to avoid conflicts can be thorough screening process that identifies the conflict of interests that exists and if they do, generate a conflict of interest waiver that ensures agreement of the client with the consulting company to eliminate their objections regarding the relationships (CNA Risk Control, 2014). Adding to that, they should make themselves independent of the external partners or affiliations that may put the interest of the concerned client at risk (Anon., ND).

Essential principles that consultants can consider managing conflicts of interests:

- The firm should have an obligation to reveal any conflict of interest at the earliest and be proactive in monitoring further developments to ensure their stance throughout he project
- The consulting company should conform to the universal ethical consulting standards and the conditions of the engagement to establish a long-term relationship with the client (Anon., ND)

Addressing the employee's personal interests that may resist to change can be difficult task as it entails isolating the reason behind what is the reason behind their reluctance to implement the transformation. The employees should try to understand first what the change is and then expect to be understood (Covey, 1989). Similarly, managers should also understand the specific changes and identify the employees that may be affected and then think about why they might resist the change. They should also hold weekly meetings and workshops to target the employees that show resistance to change and then help them to align their interests with company's vision (Quast, 2012).

Team and Skill Requirements

Expertise Requirement

ERP systems have proven to be the organisation's backbone being a core component of information management. If successfully implemented, ERP systems can enhance effectiveness of the organisation, significantly improve performance by streamlining departments that were incorporated in the transformation and establish interdepartmental cohesiveness for faster responsiveness to problems. For the engagement with Queensland health department, it is imperative for the consulting firm to have exceptional change management experience combined with efficient communication skills. This is salient for the reengineering of the client's existing financial services (Kim , et al., 2005). Since it is also possible to add onto the terms of engagement as the project goes on the team must also be open to certain changes in engagement and behaviour. They should have the internal human resources with capabilities to implement the ERP change but if they anticipate a shortfall in their internal resources, turning to external human resource could facilitate the change and advocate change more successfully (Kim , et al., 2005). Coordination among different operational areas is vital as it is the most common impediments to implementing ERP change (Cowan, 2017). This is usually the responsibility of the senior management however consultants are required to ensure the adequacy of the involvement. Just like the previous version of SAS being used at the Health Department, the SAS HANNA version might also be a subject to customisation to fit the requirements of the client. Therefore, expertise in overseeing configuration of the system to fit the requirements of the departments incumbent database and information system would also be vital (Kim , et al., 2005). The team should also be able to articulate their abilities to meticulously plan and negotiate with departments involved to devise management structures and criteria. Last but not the least, organisational leadership should be involved to advocate the vision of the ERP change across all departments to ensure smooth integration (Kim , et al., 2005).

Principal to lead the engagement

- Kelly Samson
    - Kelly is a transformation specialist who has extensive experience in organisational design and change with over 20 years of consulting experience. She has also guided change management for the merger of two government departments in Australia. Taking under consideration her experiences, she would be an ideal change management and transformation specialist to ensure smooth integration of the ERP system in finance, logistics, warehousing, assets, and procurement departments into their existing system (NOUS, n.d.).

Team Structure & Project Duration Estimate

While the principal will be responsible for leading the engagement, the managing director needs to review the case and the client's requirement analysis before responding to the RFP. A senior consultant would report to the managing director providing technical deliverables of the project with the help of three junior consultants and associates for technical support. Therefore, a team of six should be sufficient for the project. Not only would this structure of the team keep the budget under control but also can be efficient while communicating and coordinating with the department within Queensland Health; who will also provide resources to aid with the completion of the project. Considering the Queensland Health past similar projects, this project is focused on the financial system renewal which are regularly assessed based on their scope and timeline (Hendry, 2018). Nonetheless, for the initial proposal the project duration should not take longer than three years bearing in mind any contingencies along the way.

Supporting Frameworks, Tools, and Data

Consulting Frameworks and Tools

- Conceptual Research Framework
    - By identifying the four propositions as represented in the framework centred around organisation, user, ERP system and ERP vendor go hand in hand to ensure the successful implementation of the ERP system. For instance, with the LATTICE upgrade project with IBM, the system environment did not support the requirement of the organisation and ended up in a catastrophic event where many employees were left unpaid, over paid or not paid at all (Cowan, 2015). Therefore, this framework conceptualises key aspects of the engagement process to oversee the transformation by warranting the system compatibility with the organisation and its impact on the users (Zhang, et al., 2005).

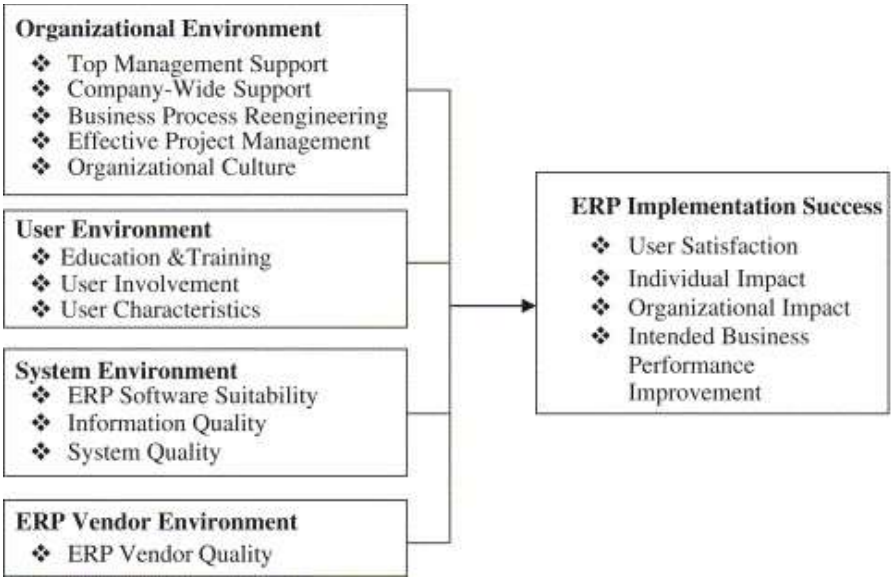

Figure 1 Conceptual research framework (Zhang, et al., 2005)

- Power-interest grid for stakeholder management
    - This can help Nous Group identify key stakeholders and prioritise them based on their power and interest in the project. With this they can further develop appropriate strategies for important stakeholder groups based on their engagement and influence on the project; for example, the Minster of Health would have significant influence on the decision making (Thompson, n.d.). Since, there are many stakeholders involved in the implementation of the ERP it is important to analyse them and establish an engagement plan based on the framework.

- Boston Consulting Group's four point to change management
    - Leaders must be involved personally to facilitate the change and lead the engagement with the client which will act as inspiration for the employees as it will advocate change (BCG-Analysis, n.d.).
    - The change must take place rapidly to avoid an engagement dip that can be discouraging for employees. This will enable the employees to see what the change will deliver and ensure their persistent involvement in the transformation (BCG-Analysis, n.d.).
    - Employees must be included via two-way communication stream to maintain transparency and to empower them to act on the organisation's vision. For the ERP system upgrade it is imperative for the employees to be involved to ensure the smooth integration into the Health department's current system (BCG-Analysis, n.d.) (Cowan, 2017).
    - Enable new capabilities and behaviour to induce and sustain the change in the long-run. This includes providing workshops and upskilling the Queensland Health's employees so that they can adapt to the change and continue to provide value added service to stakeholders (BCG-Analysis, n.d.).

Figure 2 Solutions and Tools for Change Management Success (BCG-Analysis, n.d.)

Decision on RFP Response

NOUS group's interaction with Australian Government

NOUS group have had numerous opportunities to engage with the Australian government to accelerate digital transformation. Partnering up with one of the state's government they developed a roadmap to digitally transform the programs that came under the jurisdiction of the respective department (Joffe, n.d.). This have enabled NOUS group to develop deep understanding of the Australian government and the existing systems in place unlike IBM, who misjudged the capabilities of their system to be a good fit in the Health department organisational environment (Cowan, 2015).

NOUS group's work in Health Sector

NOUS group engaged with the Victorian Department of Health and Human services to establish an operating model to develop a patient centric approach and facilitated the transformation of the organisation to the model aiming to provide better healthcare services.

NOUS group's Transformation framework

NOUS group has segregated their transformation and change management approach in order to set goals that fosters change at the organisational level while also ensuring the workforce involvement to implement change.

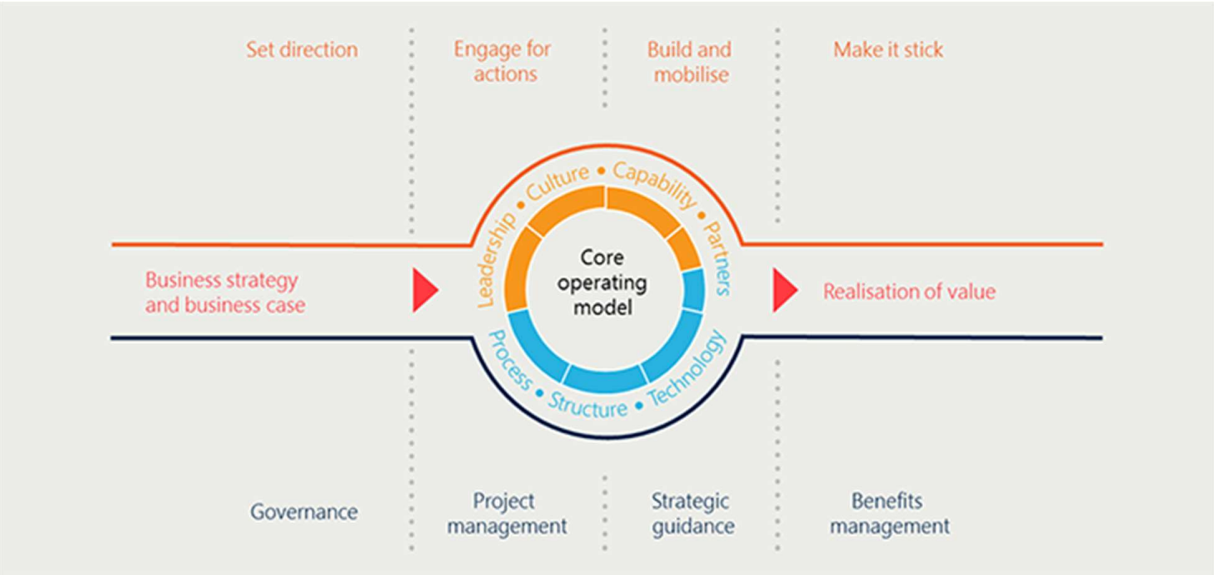

Figure 3 People: the engine of successful transformation (Samson, n.d.)

Taking under consideration the relevant experiences of NOUS group and the client's requirements it is safe to say that NOUS group should respond to the RFP. The client demands a transformation of their incumbent SAP ERP system for finance and procurement department to the cloud-based

SAP HANNA version. With the extensive background of the consulting firm in digital design and transformation across health care sectors and the public sector, NOUS group may be the ideal candidate to lead the transformation of Queensland Health ERP transformation project.

Bibliography

Anon., ND. *Conflict of Interest | International Federation of Consulting Engineers.* [Online] Available at: http://fidic.org/node/752 [Accessed March 2019].

BCG-Analysis, n.d. *Solutions and Tools for Change Management Success.* [Online] Available at: https://www.bcg.com/en-au/capabilities/change-management/solutions.aspx [Accessed March 2019].

Caldas, M. P. & Wood, Jr., T., 1999. How consultants can help organizations survive the ERP frenzy. *Academy of Management,* 6 August.

CNA Risk Control, 2014. Management Consulting – Client and Engagement Intake: Practical Considerations and Tools. In: *Ethical Analysis.* s.l.:CNA, p. 13.

Covey, S. R., 1989. *The 7 Habits of Highly Effective People: Powerful Lessons in Personal Change.* s.l.:Free Press.

Cowan, P., 2015. *Revealed: Queensland's case against IBM.* [Online] Available at: https://www.itnews.com.au/news/revealed-queenslands-case-against-ibm-399308 [Accessed March 2019].

Cowan, P., 2017. Qld Health starts work on major SAP installation. *itnews*, 23 February .

Coyne, A., 2017. *Contractor army wanted for Qld Health's middleware overhaul.* [Online] Available at: https://www.itnews.com.au/news/contractor-army-wanted-for-qld-healths-middleware-overhaul-455696 [Accessed March 2019].

Government, Q., 2017. *Queensland Government QTenders.* [Online] Available at: https://www.hpw.qld.gov.au/qtenders/tender/display/tender-details.do?CSRFNONCE=B44358E5E727C84D2341B835CACC58EF&id=20401&action=display-tender-details [Accessed 27 March 2019].

Health, Q., 2015. *Queensland Government Publications.* [Online] Available at: https://www.health.qld.gov.au/__data/assets/pdf_file/0031/442939/ehealthinvestmentstrategy.pdf [Accessed 27 March 2019].

Hendry, J., 2018. *Qld Health's new ERP system faces funding shortfall.* [Online] Available at: https://www.itnews.com.au/news/qld-healths-new-erp-system-faces-funding-shortfall-511907 [Accessed March 2019].

Joffe, G., n.d. *Accelerating digital transformation in government.* [Online] Available at: https://www.nousgroup.com/case-studies/accelerating-digital-transformation-government/ [Accessed March 2019].

Kim , Y., Lee, Z. & Gosain, S., 2005. Impediments to successful ERP implementation process. *Business Process Management Journal,* 11(2), pp. 158 - 170.

Manongdo, R., ND. *Queensland Health Payroll system.* [Online] Available at: https://www.academia.edu/8096282/Queensland_Health_Payroll_System_-_a_case_study_on_Business_Process_Management_and_application_Enterprise_Integration [Accessed 28 March 2019].

NOUS, n.d. *Kaushal Vyas | Principal.* [Online] Available at: https://www.nousgroup.com/people/kaushal-vyas/

NOUS, n.d. *Kelly Samson | Principal.* [Online] Available at: https://www.nousgroup.com/people/kelly-samson/

Pearce, R., 2017. *COMPUTERWORLD.* [Online] Available at: https://www.computerworld.com.au/article/613643/fears-over-dr-shortcomings-queensland-hospitals/ [Accessed 27 March 2019].

Purnell, F., 2017. *insideSAP.* [Online] Available at: https://insidesap.com.au/queensland-health-gears-up-for-sap-erp-transition/ [Accessed 27 March 2019].

Quast, L., 2012. *Overcome The 5 Main Reasons People Resist Change.* [Online] Available at: https://www.forbes.com/sites/lisaquast/2012/11/26/overcome-the-5-main-reasons-people-resist-change/#311a51c53efd [Accessed March 2019].

Samson, K., n.d. *People: the engine of successful transformation.* [Online] Available at: https://www.nousgroup.com/insights/people-engine-transformation/ [Accessed March 2019].

Thompson, R., n.d. *Stakeholder Analysis.* [Online] Available at: http://www.citethisforme.com/cite/sources/websiteautociteeval [Accessed March 2019].

Zhang, Z. et al., 2005. A framework of ERP systems implementation success in China: An empirical study. *International Journal of Production Economics,* 98(1), pp. 56-80.